

Functional Instability of the Ankle and the Role of Neuromuscular Control: A Comprehensive Review

Douglas H. Richie, Jr., DPM, FACFAS

A review of current knowledge of the clinical syndrome of functional ankle instability is presented. Recent evidence has demonstrated that the majority of patients with functional instability of the ankle do not have mechanical hypermobility of the ankle joint. Functional instability of the ankle results from a loss of neuromuscular control. Components of neuromuscular control include proprioception, muscle strength, muscle reaction time, and postural control. Proprioceptive deficits lead to a delay in peroneal reaction time, which appears to be a peripheral reflex. Proprioception and eversion muscle strength improve with the use of passive supportive devices. Balance and postural control of the ankle appear to be diminished after a lateral ankle sprain and can be restored through training that is mediated through central nervous mechanisms. Methods of detecting deficits in neuromuscular control are presented along with rehabilitation techniques to treat functional instability of the ankle. (The Journal of Foot & Ankle Surgery 40(4):240–251, 2001)

Key words: ankle instability, ankle proprioception, ankle sprains, ankle sprain rehabilitation

Ankle sprains are the most common injury affecting athletes, accounting for up to 25% of all lost time from participation in sports (1, 2). Lateral ankle ligament sprains comprise 85% of all ankle sprains while eversion sprains of the deltoid ligament comprise 5% of sprains and syndesmosis sprains comprise 10% of these injuries (3–5).

Despite ongoing research and newer technologies for treatment interventions, patients suffering a lateral ankle sprain have a significant chance of developing long-term sequela. After a single lateral ankle sprain, it has been estimated that 70–80% of athletes will suffer a recurrent sprain (6, 7). Symptoms of residual instability develop in 20–40% of patients after a lateral ankle sprain (8–10). Long-standing lateral instability can predispose to osteoarthritis (11).

Studies have shown that over half of patients with chronic lateral ankle instability do not have clinical or radiographic evidence of mechanical instability with standard anterior

drawer stress or inversion stress (12, 13). At the same time, mechanical instability can be demonstrated in patients shortly after a first-time ankle sprain, and this instability will disappear in just 12 weeks of rehabilitation (14).

Surgical procedures for the chronic unstable ankle are designed to repair, augment, or replace injured ligamentous structures (15–24). Yet, less than half of patients with functional ankle instability do not have anatomic deficiencies in these ligaments. Thus, surgical procedures cannot be offered as a remedy to patients with functional ankle instability.

This article reviews current knowledge about functional ankle instability beginning with an analysis of the pathomechanics of the lateral ankle sprain. Next, the components of neuromuscular control over stability are examined, including an analysis of the role of passive supportive devices to augment dynamic defense mechanisms. Finally, a clinical assessment program is proposed that should be an essential component of the presurgical evaluation of any patient with chronic instability of the ankle.

Mechanical Versus Functional Instability

Mechanical instability involves an anatomic abnormality such as disruption of one or more lateral collateral

Address correspondence to: Douglas H. Richie, Jr., DPM, FACFAS, 550 Pacific Coast Hwy., Suite 209, Seal Beach, CA 90740.

Received for publication August 17, 2000; accepted in revised form for publication April 15, 2001.

The Journal of Foot & Ankle Surgery 1067-2516/01/4004-0240\$4.00/0
Copyright © 2001 by the American College of Foot and Ankle Surgeons

ligaments of the ankle (25). Biomechanical deformities such as tibial varum, rearfoot varus, or forefoot valgus will create mechanical instability by setting up compensation mechanisms that induce supination moment to the talocrural joint (26, 27). Congenital ligamentous laxity can also contribute to mechanical joint hypermobility (13).

Freeman first described functional instability in 1965 as a condition in which a patient has "recurrent sprains and/or a feeling of giving way of the ankle" (28). Konradsen attributed the cause of functional instability to both mechanical and functional causes in stating that functional instability results from "damage to mechanoreceptors in the lateral ligaments or muscle/tendons with subsequent partial de-afferentiation of the proprioceptive reflex" (29). Mechanical instability can cause functional instability (14). Damage to the sensorimotor control system of the ankle has been implicated as a primary cause of recurrent inversion injuries (25, 28, 30, 31).

Not all ankles that are mechanically unstable actually function in an unstable fashion (14). In a study of 444 soccer players, Tropp found that 128 players had functional instability of one or both ankle joints (12). Of the 159 functionally unstable ankles, 66 (42%) were found to be mechanically unstable (Fig. 1). In their study of 117 functionally unstable ankles, Vaes et al. demonstrated mechanical instability in only 41 ankles (supine stress talar tilt greater than 7°) (13). Staples has also questioned cause-effect relationship between functional and mechanical instability (32). A summary of the contributory factors of functional ankle instability is provided in Figure 2.

Pathomechanics of the Lateral Ankle Sprain

Most inversion ankle sprains occur when the foot is plantarflexed at the ankle and a supination moment force is applied to the foot while an external rotation force is applied to the leg (33, 34). The plantarflexed ankle is most vulnerable to injury because: 1) The plantarflexed talus

with its narrow posterior body is thrust forward between the malleoli and has less stability than in its dorsiflexed position, and 2) the anterior talofibular ligament and the calcaneofibular ligament are under greatest mechanical strain in a plantarflexed position (35).

Clinical situations commonly attributed to precipitating an ankle sprain include landing from a fall, landing from a jump on an opponent's shoe, stepping off a curb, and stepping down stairs (1-5, 33, 34). In each of these situations, forefoot contact would precede rearfoot (calcaneal) contact, and the forefoot through a gearing mechanism and movement coupling sequence would transmit a supination moment to the talus and calcaneus.

The fact that the pedal joints are primarily responsible for transmitting supination moment to the ankle joint raises concern about simultaneous injury to the ligaments of the foot when a patient suffers a lateral ankle sprain. Indeed, it has been documented that there is a subpopulation of patients with a history of lateral ankle sprains that demonstrate mechanical instability of both the ankle joint and subtalar joints (36-40). In their study of 12 subjects with a history of unilateral ankle sprains compared to eight healthy controls, Hertel and co-workers demonstrated that 78% of patients with excessive talar tilt documented with fluoroscopy also demonstrated laxity of the subtalar joint with manual testing, and 67% demonstrated talocalcaneal laxity of the subtalar joint under stress fluoroscopy (41).

The rotational force applied via the subtalar and mid-tarsal joints to the talocrural joint has much greater potential for creating ligamentous injury of the ankle than does pure inversion and anterior drawer stress commonly employed with clinical testing and experimental models. A clear indication of this overlooked fact is found in studies of the inherent osseous stability of the talocrural joint. In one study by Fraser and Ahmed (42) and another study by McCullough and Burge (43), cadaver models demonstrated that, with increased axial load, rotation in the talocrural joint decreased. With intact ligaments and after ligament release, increased axial load was associated with decreased axial rotation of the talus within the mortise. However, both of these studies did not permit normal subtalar motion to occur when axial load was applied. Stormont et al. allowed subtalar joint motion in their study of 21 cadaver specimens with application of an axial load of 150 pounds (44). After serial sectioning of the ankle ligaments, Stormont and co-workers concluded that the articular surfaces of the talocrural joint account for 30% of the stability in rotation and 100% of the stability with inversion force resistance (44).

Cass and Settles noted that Stormont's study would not allow rotation of the foot and ankle along with inversion and eversion (45). Either rotation or inversion-eversion was constrained. Cass and Settles constructed an experimental model that would allow both ankle and subtalar

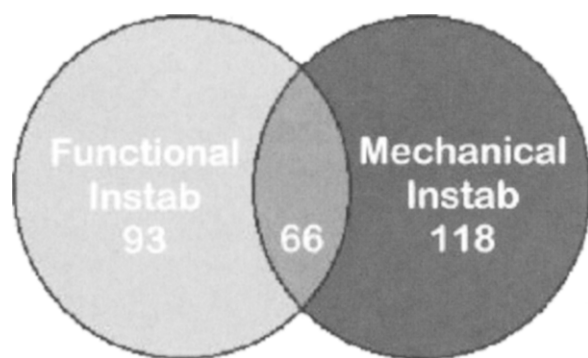


FIGURE 1 The association between functional and mechanical instability of the ankle joints in 444 soccer players. (Reprinted, with permission, from Tropp, H., et al. *Int. J. Sports Med.* 6:180, 1985.)

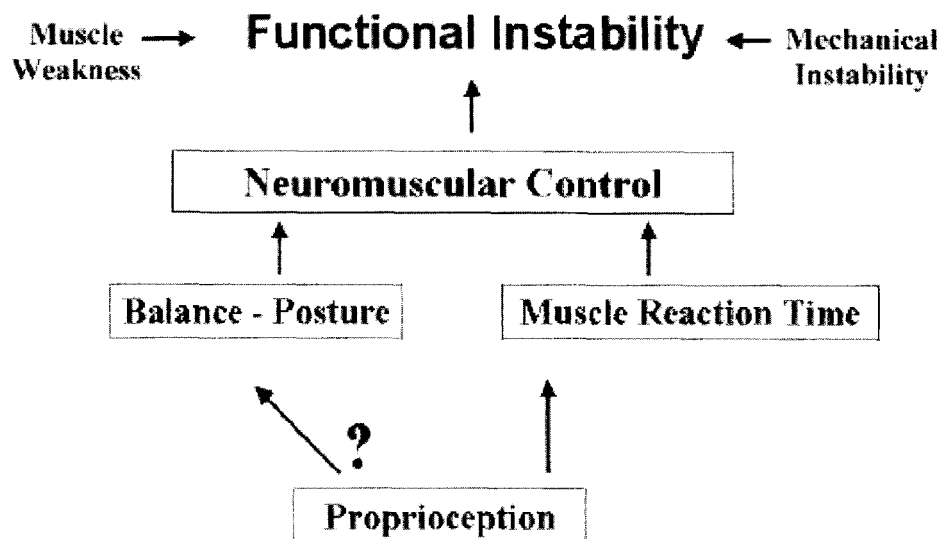


FIGURE 2 A summary of the contributory factors of functional ankle instability.

joint motion about all three reference axes of rotation (inversion/eversion, dorsiflexion/plantarflexion, adduction/abduction). They repeated Stormont's study utilizing 19 cadaver specimens where axial load was applied while axial rotation was unconstrained. In every specimen, significant talar tilt did not occur until after both ligaments were released, averaging 20.6°. External rotation of the leg increased by 4.9° after anterior talofibular ligament sectioning and 12.8° after sectioning of both the anterior talofibular and calcaneofibular ligaments. These investigators concluded that the anterior talofibular ligament and the calcaneofibular ligament work in tandem to stabilize the talocrural joint while the articular surfaces appear to provide minimal restraint in preventing tilting of the talus in the ankle mortise.

These cadaver studies were all performed on axial loaded cadaver models where the foot was in a neutral (90°) position to the leg. Anatomically, the ankle is in its most stable osseous position when it is dorsiflexed and in its most unstable osseous configuration when it is plantarflexed (34). While the plantarflexed ankle lacks osseous stability, this joint position actually improves neuromuscular stabilization of the ankle.

Neuromuscular Control: The Role of Proprioception

Considerable research has been conducted on the compensatory response to humans to perturbations occurring during gait (46–48). The rapid response of lower extremity musculature to such perturbations has been measured on tilting platforms and treadmills during both static stance and gait (14, 29, 46, 49, 50). The reflex mechanism of the body as a whole attempts to fulfill

one primary requirement: to maintain the body's center of mass over the feet.

Sudden displacement of the foot or feet activates a sequence of muscle firing that is dependent upon central generators and programs interacting with peripheral reflexes (51). Afferent information is provided by proprioceptive, visual, vestibular, and auditory systems. The programmed leg muscle response varies according to the direction of perturbation and the phase of gait or alignment of the foot on the ground. Unilateral foot and leg displacement evoke a bilateral response pattern with a similar latency of onset on both sides (approximately 55 ms) (52). Both proximal and distal activation of muscles occur in the legs with activation of agonist and antagonist groups. In cases of unilateral displacement of the foot (sprain), a rapid contraction of the muscles of the contralateral occurs to provide a stable base of support (53). The mediation of afferent input and coordination of bilateral response is controlled by spinal interneuronal circuits, which themselves are under control of the central nervous system (51).

Proprioceptive afferent input is provided by receptors located in muscles, tendons, joints, and other tissues. The monosynaptic stretch reflex involves muscle spindle receptors connecting I-a nerve fibers as well as Golgi tendon organs connecting to I-b fibers (54). During rapid perturbation such as tripping or falling, monosynaptic reflexes are absent and compensation occurs as a result of transmission along group II and III afferent fibers from secondary muscle spindles (52). These connect through a polysynaptic reflex system to generate an appropriate response. A central program as well as supraspinal influences interacts in a complex manner which is poorly understood. The contribution of vestibular and visual input

to these reflexes is minimal (51). Gravity and pressure on the joints and on the plantar skin surface of the feet may be critical to these reflexes (55). Experiments during weightlessness in space and during emersion in water show a compromise in peripheral afferent input and increased reliance on visual and vestibular cues (51). Otherwise, the vestibular system is primarily involved during falls with stabilizing the head and, along with visual input, compensating body sway (49). At the same time, leg muscle EMG activity occurring after a fall is predominantly induced by proprioceptive reflex rather than vestibular input (51).

Several studies have shown that ankle ligamentous injury will lead to distortion of joint position sense. Glen-cross and Thornton found significant differences between sprained and nonsprained ankles in an active positioning task (31). Konradsen studied 44 patients with clinical grade II to III first-time ankle inversion sprains (14). A significant loss of ankle joint position sense was found in the sprained ankle that persisted 12 weeks after injury. Interestingly, Konradsen found no increased peroneal reaction time on the injured side compared to the noninjured side at 3, 6, and 12 weeks after injury. In comparing to previous work performed on subjects with chronic ankle instability, Konradsen concluded that a first-time ankle sprain does not compromise peroneal reaction time.

There is some debate about the true location of the proprioceptor organs necessary to maintain ankle joint stability. Freeman et al. have demonstrated that numerous articular nerve fibers can be found in mechanoreceptors of the capsules and ligaments of the ankle joint (56). They speculated that capsular and ligamentous injuries lead to partial or permanent joint de-afferentiation. Lack of sensory input from these joint mechanoreceptors would theoretically lead to functional ankle instability. However, several investigations have demonstrated that ankle joint proprioception may not depend on ligament or capsule mechanoreceptors. DiCarlo and Talbot evaluated ankle joint proprioception following anesthetic injection of the anterior talofibular ligament (57). They found that bilateral balancing ability actually improved after anesthesia. Feuerbach et al. examined 12 noninjured subjects before and after anesthesia applied to the anterior talofibular and calcaneofibular ligaments (58). No significant differences could be found between accuracy of ankle positioning tasks between the nonanesthetized and anesthetized conditions.

Afferent feedback for ankle joint proprioception may be provided from the skin, muscle, and other joint receptors (59–61). Robbins has demonstrated that cutaneous and pressure receptors located on the plantar surface of the foot have significant influence over protective activation of the lower leg musculature (55).

Neuromuscular Control: Muscle Reaction Time

In terms of lateral foot and ankle perturbations, several studies have evaluated the timing and power of neuromuscular response in the lower leg muscles. Konradsen tested 10 subjects with mechanically stable ankles walking and standing on a trap door apparatus capable of suddenly inverting the foot 30° in the frontal plane (62). Peroneal latency (time for initial peroneal EMG activity) was 54 ms. This reflex latency was significantly faster when the foot was placed in an inverted position rather than an everted position prior to sudden inversion on the trapdoor.

A significant faster reflex time was found in the peroneal muscles compared with the quadriceps and hamstring, leading Konradsen to conclude that these later muscle reflexes rely on a centrally coordinated program. The peroneal reflex appeared more reliant on peripheral receptors in the tendons or muscles because of the shortened reaction time occurring when the foot was placed in an inverted position rather than everted position.

Konradsen also studied electromechanical delay, which is the time necessary to generate eversion moment after initial EMG activity is noted in the peroneal musculature. This delay of 72 ms must be added to the peroneal reaction time of 54 ms before sufficient muscle tension can be developed to prevent inversion. At least 126 ms will thus pass before true protective muscle activity develops, which is too long of a delay before the ankle can invert to the point of ligament failure. In this study, it took only 80 ms for subjects to invert 30° on a trapdoor. Isakov and Nawoczenski presented this concern about peroneal reaction delay (30, 50).

Protective activation of the peroneal muscles (peroneal reaction time) has shown significant delay in patients with chronic lateral ankle instability. Konradsen and Ravn found a significant mean delay of 17 ms in peroneal reaction time in patients with unstable ankles compared to patients with stable ankles (29). Brunt and co-workers found a 13-m difference in patients with previous grade II ankle sprains compared to healthy subjects (46).

Karlsson studied peroneal reaction times in subjects with unstable chronic lateral ankle instability (63). The reaction time was significantly shorter in stable ankles compared to unstable ankles (68.8 ms vs. 84.5 ms, $p < .001$). Interestingly, when the unstable ankles were taped, the reaction time improved significantly ($p < .05$). Not all ankles responded the same to tape; the most unstable ankles responded best. This interaction between passive external supports augmenting the dynamic defense mechanism has been extensively studied (62–66).

A contradiction to these findings can be found in two separate studies published by Isakov and Nawoczenski (30, 50). Both studies failed to show a significant decrease in peroneal reaction time in patients with

sprained ankles versus uninjured patients. However, these studies used unilateral inversion stress conditions, while studies by Karlsson and Brunt used bilateral perturbation on tilt platforms.

Experimental research suggests that the peroneal latency added to electromechanical delay in the peroneal musculature does not appear to be capable of protecting the ankle from sudden inversion force when tripping, falling, or landing on uneven terrain. In evaluating previous research, this delay would exceed 150 ms (30, 62). However, these tests were conducted on subjects in a standing position with lower extremity muscles at rest. In the walking, running, or jumping conditions, humans demonstrate preactivation of lower extremity muscles prior to foot touchdown (67, 68). Preactivation of lower leg muscles prior to ground contact increases segmental reflex activity and stretch velocity (69–71). Preactivated peroneal muscles, with fully activated cross-bridges of contractile units prior to foot touchdown, would provide significantly greater muscle force upon touch down without significant time delay (72). A higher rate of tension rise will occur during stretching as the foot inverted due to eccentric lengthening contractions that occur in the peroneal musculature. The force per active fiber ratio is greater during eccentric muscular contractures than during concentric conditions (73).

Plyometric contractions involve a stretch-shortening sequence which combines eccentric and concentric contractions (74). The force developed from stretch-shortening is greater than in an isometric contraction (75). In normal running, stretch-shortening determines muscle stiffness and accounts for the spring-like elastic properties of muscle during landing, push-off, and acceleration of the body (72). Higher brain centers may regulate muscle stiffness prior to touchdown when perceiving changes of terrain or surface hardness (68, 76).

Vestibular and auditory cues may be linked to precontraction of lower leg muscles during tripping and falling (77). Melvill-Jones and Watt demonstrated that human subjects deprived of visual input when dropped from a height required a minimum of 74 ms to activate lower leg muscles and prepare for impact (78). Effective buildup of muscle tension could not occur until at least 102 ms. Falls from heights under 5 cm, occurring in less than 100 ms, resulted in insufficient activation of lower leg musculature in human subjects. Thus, Melvill-Jones actually measured sharper impact force on the feet of human subjects falling less than 10 cm compared to those falling from heights greater than 17 cm. Falls above 18 cm, taking 190 ms, were required to fully activate a protective shock-absorbing reaction. Some protection is formed from reflex occurring in the otolith apparatus, but requires a height of at least 7.5 cm to activate this reflex.

Winter has shown that the foot passes as close as 5 mm to the ground during the swing phase of walking gait (79).

Higher impact force applied to the ankle has greater potential to cause an ankle sprain. A force of one body weight (BW) applied more than 3.4 cm medial to the midline will cause an ankle sprain (80). A four-BW force need only be applied to 0.85 cm medial to the midline for a sprain to occur.

From their data on muscle eversion power in an isometric condition, Ashton-Miller and co-workers calculated the potential effect of precontracted muscle action prior to ground contact on a 15° inverted surface (80). The resulting eccentric contraction would increase muscle force from 35.8 to 68.0 N/m. Total equivalent muscle force based on the lever arm of the peroneal longus and brevis was calculated at 2533 N, enough to rupture or tear the peroneal tendons or avulse the styloid process of the fifth metatarsal base.

Effects of Passive Support

Augmentation of the dynamic defense system to prevent lateral ankle sprains can be accomplished in two ways: passive supportive devices and specialized balancing training to augment neuromuscular control. Many studies have been conducted on the effects of tape and braces to prevent and treat ankle sprains. Athletic tape has been reported to effectively prevent ankle sprains in athletes (64, 81–84). Other studies have shown either no reduction of injury (85) or a hindrance of athletic performance with taping of the ankles (86–88). Until recently, most studies on the effectiveness of tape to resist ankle inversion were carried out in nonweightbearing conditions (82, 89–93).

Manfroy and co-workers studied the effects of tape and prewrap on healthy subjects under weightbearing conditions and found that tape significantly improved ankle eversion moment (resistance to inversion) (65). However, after 40 minutes of exercise, this protective benefit of tape was lost. Other studies have verified that tape can lose its mechanical strength as soon as 10 minutes after exercise (64, 93–96).

The effect of footwear in providing passive support for the ankle has been studied. Garrick not only found protective benefit from taping but also a lowering of risk when athletes combined tape with a high-top shoe (5). Rovere found that a lace-up brace was more effective than tape in preventing ankle sprains in collegiate football players and that combining a brace with a low-top shoe was better than a high-top shoe (97).

The effectiveness of tape may not be dependent on its mechanical effect on the ankle. Karlsson studied 20 patients with mechanically unstable ankles verified by stress radiography utilizing the Telos device (63).

When these ankles were taped, no reduction of instability could be measured with stress radiography. However, peroneal reaction time was significantly shortened when the unstable ankles were taped. Overall, peroneal reaction time was significantly shorter in stable versus unstable ankles. Karlsson concluded that tape helps patients with unstable ankles by facilitating proprioceptive and skin sensory input to the central nervous system. Glick also found that tape improved peroneal reaction time in patients with significant talar tilt (64). In patients with stable ankles, tape caused no improvement of peroneal reaction time.

As with previous studies on tape, studies on the use of ankle braces have been primarily performed on nonweightbearing subjects or on cadaver models that had the foot disarticulated. Greene studied a semirigid ankle brace in nonweightbearing human subjects to determine passive restriction of ankle joint range of motion (90). In this study, the semirigid orthosis was more effective than the tape in limiting inversion, both during and after exercise. Gross has published three studies testing various ankle braces utilizing The Biodex Stability System^{®1} device to measure passive inversion or eversion in nonweightbearing human subjects (91, 92, 98). Shapiro studied five cadaver ankles to determine the effects of taping and bracing (60). A Materials Test Systems device determined passive inversion force, moment, and stiffness. The cadaver specimens were mounted on a platform with only the calcaneus contacting and with the forefoot removed at the tarsometatarsal joints. The braces and tape provided a more than two times improvement of resistance to inversion.

Ashton-Miller and Manfroy designed a special testing apparatus to measure functional eversion ankle strength of human subjects under full weightbearing conditions in a neutral position and at 32° plantarflexion (65, 80). Ashton-Miller utilized this unipedal strength test to measure maximal voluntary resistance to inversion moment developed by 20 healthy adult men in various conditions, including with ankle taping in place, or any one of three different ankle braces (80). A 3/4 high-top shoe increased eversion strength significantly at 0° plantarflexion (5.9 N-m increase) and at 32° plantarflexion (3.3 N-m increase). Ankle taping or bracing increased inversion resistance significantly, although no differences were found between taping and any of the three braces. In any shoe, ankle taping or bracing improved inversion resistance by 7.8% at 0° plantarflexion and by 4.6% at 32° plantarflexion.

Of interest in this study was the finding that increased muscular eversion moment developed as the ankle was more plantarflexed, yet effectiveness of passive support

provided by ankle braces decreased as the ankle was plantarflexed. Ashton-Miller calculated that at 15° inversion, the ankle eversion muscles isometrically developed an eversion moment up to six times larger than that developed by a 3/4 high athletic shoe alone (80). This eversion muscle strength was also three times larger than that developed passively by tape or any one of three popular ankle braces. In a plantarflexed, inverted ankle, the activated and powerful contraction of the peroneal muscles provides a dynamic defense mechanism that is far more effective than any combination of footwear, taping, or bracing.

This dynamic defense mechanism appears most effective when the ankle is in its most vulnerable position: plantarflexed and inverted. Peroneal latency is significantly shorter when the ankle is placed in inversion compared to eversion (62). Active tension in the peroneal musculature is more than 63% greater in an inverted ankle compared to a neutral ankle (80). When the ankle is plantarflexed 32°, the peroneal muscles generate 73% more power than in a neutral ankle position (80). These findings suggest a length-tension phenomenon as well as an augmentation of the stretch reflex.

Coordination, Balance, and Postural Control

Loss of proprioception may lead to lengthened peroneal reaction time as well as impaired balance on the supportive limb (99, 100). Loss of balance and postural control can be measured objectively with stabilometry, involving calculation of center of pressure on a force platform. With stabilometry evaluation, subjects are required to stand on one leg with the arms folded across the chest in a modified Robert position. The force platform detects anteroposterior and medial lateral shifts in the center of mass as it is maintained over the supportive foot.

Single-leg balance, as measured by stabilometry, is mediated by joint proprioception. Since balance assessment and training is performed in a weightbearing position, it may be more representative of lower extremity proprioception ability than other nonweightbearing assessments (101, 102). Cornwall used stabilometry to measure postural sway during single-leg stance in 20 subjects with a history of inversion ankle sprain (99). These subjects showed a significant impairment of single-leg stance compared to a control group.

Tropp et al. studied 47 male soccer players and found that those players with functional ankle instability had significantly higher stabilometry scores, indicating decreased balance control, than players without functional instability (12). This same study showed no difference in postural balance, as measured by stabilometry, between players with mechanically unstable ankles and those without. Therefore, loss of postural control and balance appears to be more closely associated with functional

¹ Biodex, Shirley, NY.

instability of the ankle than mechanical instability. Previously, Tropp had shown a predilection for ankle sprains among soccer players with abnormal stabilometry findings (103).

Restoration of balance and proprioception is the cornerstone of rehabilitation programs for the functionally unstable ankle. These programs often employ unstable balancing platforms or disks that require training on a weightbearing limb (12, 104, 105). Several studies have shown that disk or wobble board training will improve coordination, balance, strength, and proprioception, while reducing overall functional instability of the ankle.

Tropp et al. placed 65 male soccer players with a history of ankle instability into a 10-week ankle disk coordination program (106). This group was followed and compared to a control group for 6 months of soccer competition and practice. Ankle disk training reduced the risk of injury in previously injured players to a level of those without any previous injury. Also, ankle disk training prevented ankle injury at the same level as did an ankle orthosis.

Tropp and Gillquist have shown that ankle disk training can improve stabilometry scores and reduce symptoms of functional instability (105, 107). Gauffin et al. studied 10 male soccer players with functional instability of the ankle, before and after 8 weeks of ankle disk training (108). Stabilometry recordings were made with a force platform and optoelectronic movement recording system. A significant decrease in postural sway occurred after ankle disk training. The training program achieved supranormal values and even improved scores on the untrained limb. The authors concluded that these findings validated the theory that balance relies on central motor programs. Pathologic stabilometric values reflecting functional ankle instability do not appear to be the result of peripheral proprioceptor deficits (99, 108).

Rozzi and co-workers offered further challenge to the proprioceptive deficit theory (109). Twenty-six subjects were divided into two groups: one experimental group with self-reported functional ankle instability and one group without functional instability (13 subjects in each group). The Biodex Stability System[®] was utilized as an assessment and training. The experimental group had significantly impaired balance ability at the beginning of the study. However, there was no difference between the groups after training three times a week for 4 weeks. A deficit imbalance ability was also found in the uninvolved limb. Also, a balance score improvement was found in the untrained limb. These findings also suggest a centrally mediated neuromuscular mechanism responsible for the maintenance of balance and posture.

Muscle weakness has been cited as a causative factor of chronic ankle instability. However, data justifying this concern have been misleading. Bosien found peroneal

muscle weakness in 23 of 35 ankles (66%) using manual muscle testing techniques (110). In a long-term clinical study by Staples, 9 of 21 (43%) patients demonstrated residual weakness of the peroneal muscles with manual muscle testing (111). This subjective assessment of peroneal muscle weakness has not been verified with objective measurements taken of subjects with chronically unstable ankles after grade II or grade III ankle sprains.

Lentell and co-workers measured peak torque of inversion and eversion on 33 subjects with chronic lateral ankle instability utilizing objective isometric and isokinetic techniques (104). No significant differences were found in muscle strength between the involved and uninvolved ankles. However, balance testing of these same subjects with a modified Romberg test showed abnormalities in 55% of the subjects.

Konradsen et al. measured isometric eccentric ankle eversion strength in 44 patients at 3 weeks and 12 weeks after a first-time grade II or grade III ankle sprain (14). At 3 weeks postinjury, eversion strength was significantly less on the injured side. However, strength had normalized by 12 weeks postinjury with no significant difference between the injured and uninjured sides.

Both of these studies measured eversion strength in an open-chain environment. Further research on functional ankle instability in a closed-chain situation is needed to determine the role of muscle weakness in the overall syndrome of functional ankle instability. In terms of neuromuscular control of the ankle, proprioception, coordination, balance, and posture control have consistently demonstrated a more crucial role than peroneal muscle weakness.

Clinical Recommendations for Assessment

Although many experimental models have been presented to evaluate functional instability of the ankle, several simple techniques can be utilized in the clinical setting to detect the presence of functional instability and allow monitoring of progression of rehabilitation programs. Clinical assessment of functional ankle instability falls into two areas: subjective historical information and clinical testing of the patient.

A patient history of recurrent sprains and/or a feeling of "giving way" of the ankle is a consistent finding among people with functional ankle instability (28). The ability to recover from these sprains can allow a further subclassification of this instability. Vaes proposed a classification system of people with functional ankle instability as being either compensated or noncompensated (13). Noncompensated ankles remain swollen for a minimum of 5 days after each sprain with maintenance of feeling of instability afterward.

Ankle Joint Functional Assessment Tool

To questions 1–9, assign a score (0–4) based on the following scale:

- 4 = much less than the other ankle
- 3 = slightly less than the other ankle
- 2 = equal in amount to the other ankle
- 1 = slightly more than the other ankle
- 0 = much more than the other ankle

Assign a score (0–4) to each of the following 9 questions:

- 1) How would you describe the level of pain you experienced in your ankle?
- 2) How would you describe any swelling of your ankle?
- 3) How would you describe the stability of your ankle when walking on uneven surfaces?
- 4) How would you describe the overall feeling of stability of your ankle?
- 5) How would you describe the overall feeling of strength of your ankle?
- 6) How would you describe your ankle's stability when you descend stairs?
- 7) How would you describe your ankle's stability when you jog?
- 8) How would you describe your ankle's ability to "cut" or change directions when running?
- 9) How would you describe the overall activity level of your ankle?
- 10) Which statement best describes your ability to sense your ankle beginning to "roll over?"

- 0 = much later than the other ankle
- 1 = slightly later than the other ankle
- 2 = at the same time as the other ankle
- 3 = slightly sooner than the other ankle
- 4 = much sooner than the other ankle

- 11) Compared with your other ankle, which statement best describes your ability to respond to your ankle beginning to "roll over?"

- 0 = much later than the other ankle
- 1 = slightly later than the other ankle
- 2 = at the same time as the other ankle
- 3 = slightly sooner than the other ankle
- 4 = much sooner than the other ankle

- 12) Following a typical incident of your ankle "rolling," which statement best describes the time required to return to activity?

- 0 = more than 2 days
 - 1 = 1–2 days
 - 2 = more than 1 hour and less than 1 day
 - 3 = 15 minutes to 1 hour
 - 4 = almost immediately
-

FIGURE 3 Ankle joint functional assessment tool. (Adapted from Rozzi, S. L., et al. *J. Orthop. Sports Phys. Ther.* 8:478–486, 1999.)

Rozzi et al. developed a subjective ankle joint functional assessment tool questionnaire (109). This questionnaire involves 12 questions previously used to evaluate the functional level of the knee joint (Fig. 3). A maximal score of 48 is used to assess stability of the ankle joint before, during, or after completion of a rehabilitation program.

Qualitative testing for functional ankle instability according to most authorities involve a modified Romberg test, a single leg-hop test, and demonstration of

proficiency with single-leg balance on a disk or wobble board. The modified Romberg test involves single-leg stance on the affected foot and leg with the other leg flexed at the knee (12, 28). The arms are crossed over the chest. The patient focuses the eyes straight ahead. Balance and stability are assessed with the eyes open and then closed. Functional instability will impair single-leg balance while stable individuals should be able to stand for 30 s without falling to the side (103, 112).

Ankle Performance Test Protocol

Subjective:

1. Current symptoms: none, mild, moderate, severe
2. Can you walk normally? (yes/no)
3. Can you run normally? (yes/no)

Performance Tests:

- Walking down stairs
- Rising on heels
- Rising on toes
- Balance on square beam
- single leg

Clinical Measures:

- Dorsiflexion range of motion
 - Anterior drawer (manual)
-

FIGURE 4 Ankle performance test protocol. (Adapted from Kaikkonen, A., et al. *Am. J. Sports Med.* 22(4):462–469, 1994.)

The single leg-hop is a modification of the Romberg test and evaluates strength, balance, and level of pain in the affected ankle (112). The patient is asked to rise on the forefoot of the single weightbearing foot and then perform five single leg-hopping maneuvers onto the forefoot in succession. The arms are outstretched to assist in balance. Pain or weakness may impair the ability to complete the task.

Balance on an ankle disk or wobble board can be an assessment and training tool for functional ankle instability (58, 106, 108, 109, 113). Cooper provided an overview of an ankle rehabilitation program using disk training (114). An ankle disk or wobble board should be available in the clinic to assess the ability of a patient to stand on a single foot while balancing the board for a minimum of 30 s, both barefoot and with shoes.

Kaikkonen et al. developed a performance test protocol to evaluate recovery after surgical repair of grade III lateral ankle ligament injuries (115). Although the scoring scale proposed was recommended for investigative studies, the protocol can be easily adapted in a clinical setting to evaluate functional stability of the ankle joint (Fig. 4). Of the four clinical tests, the ability of a patient to walk down stairs proved most reliable in placing patients in four subgroups of recovery from ankle injury (excellent, good, fair, poor).

Summary

Experimental and clinical studies of patients with functional instability of the ankle have provided considerable insight into the understanding of this complex clinical disorder. Although further research is required to improve specific clinical recommendations for evaluation and treatment, the following insights appear to have scientific validation:

1. The relationship between mechanical instability and functional instability of the ankle remains unclear.

The majority of patients with functional ankle instability do not have demonstrable evidence of mechanical instability and would not be candidates for surgical reconstruction or repair.

2. Functional instability of the ankle represents a loss of neuromuscular control. Components of neuromuscular control include proprioception, muscle weakness, muscle reaction time, and posture control.
3. Proprioceptive input from the ankle appears to originate in skin, muscle, and tendon receptors rather than ligamentous mechanoreceptors.
4. Muscle weakness has not been consistently and objectively measured in patients with functional instability of the ankle.
5. Subjects with functional instability of the ankle have demonstrated delayed peroneal reaction time compared to subjects with stable ankles.
6. The dynamic defense mechanism involves proprioceptive input of inversion moment coupled with neuromuscular activation to develop eversion moment of the talocrural joint. This defense mechanism is not capable of protecting an ankle at rest from sudden inversion perturbation. However, preactivated, trained musculature of the lower extremity will shorten electromechanical delay and allow a powerful plyometric contraction to resist significant inversion force.
7. Passive supportive devices, including braces and tape, augment proprioceptive input and reduce peroneal reaction time. These passive supportive devices also increase isometric eversion strength and have demonstrated a preventive influence over the occurrence of ankle sprain.
8. The peroneal musculature can develop a 5-fold greater eversion moment to protect the ankle from inversion sprain than any shoe, tape, or ankle brace.
9. Balance and posture control, as measured by stabilometry, appear to be a valid discriminating factor for identifying patients with functional instability of the ankle. Coordination, balance, and posture control of the ankle appear mediated by central nervous mechanisms, while proprioception appears to rely on peripheral reflexes.
10. Patients with functional ankle instability can be identified by careful history and ability to perform balance and proprioceptive tasks. Ankle disk training appears to be the most effective rehabilitation tool to reduce symptoms of functional instability of the ankle.

References

1. Mack, R. P. Ankle injuries in athletes. *Clin. Sports Med.* 1:71–84, 1980.
2. Garrick, J. G. Epidemiologic perspective. *Clin. Sports Med.* 1:13–18, 1982.

3. Baldwin, F. C., Tetzlaff, J. Historical perspectives on injuries of the ligaments of the ankle. *Clin. Sports Med.* 1:3–13, 1982.
4. Jackson, D.W., Ashley, R. L., Powell, J. W. Ankle sprains in young athletes. Relation of severity and disability. *Clin. Orthop.* 101:201–215, 1974.
5. Garrick, J. G., Requa, R. F. Role of external support in the prevention of ankle sprains. *Med. Sci. Sports* 5(3):200–203, 1973.
6. Smith, R. W., Reischl, S. F. Treatment of ankle sprains in young athletes. *Am. J. Sports Med.* 14:465–471, 1986.
7. Yeung, M. S., Chan, C. H., Yuan, W.Y. An epidemiological survey on ankle sprain. *Br. J. Sports Med.* 28:112–116, 1994.
8. Itay, S., Ganel, A., Horosowski, H., Farine, I. Clinical and functional status following lateral ankle sprains. *Orthop. Rev.* 11:73–76, 1982.
9. Bosien, W. R., Staples, O. S., Russell, S. W. Residual disability following acute ankle sprain. *J. Bone Joint Surg.* 37-A:1237–1243, 1955.
10. Brand, R. L., Black, H. M., Cox, J. S. The natural history of the inadequately treated ankle sprain. *Am. J. Sports Med.* 5:248–249, 1977.
11. Harrington, D. C. Degenerative arthritis of the ankle secondary to long-standing lateral ligament instability. *J. Bone Joint Surg.* 61-A:354–461, 1979.
12. Tropp, H., Odenrick, P., Gillquist, J. Stabilometry recordings in functional and mechanical instability of the ankle joint. *Int. J. Sports Med.* 6:180–182, 1985.
13. Vaes, P.H., Duquet, W., Pierre-Powel, C., et al. Static and dynamic roentgenographic analysis of stability in braced and non-braced stable and functionally unstable ankles. *Am. J. Sports Med.* 26:692, 1998.
14. Konradsen, L., Olesen, S., Hansen, H. Ankle sensorimotor control and eversion strength after acute ankle inversion injuries. *Am. J. Sports Med.* 26:72–77, 1998.
15. Ahlgren, O. H., Larsson, S. Reconstruction for lateral ligament injuries of the ankle. *J. Bone Joint Surg.* 71-B(2):300–303, 1989.
16. Brostrom, L. Sprained ankles: VI. Surgical treatment of “chronic” ligament ruptures. *Acta Chir. Scand.* 132:551–565, 1966.
17. Gould, N., Seligson, D., Gassman, J. Early and late repair of lateral ligament of the ankle. *Foot Ankle* 1(2):84–89, 1980.
18. Karlsson, J., Bergsten, T., Lansinger, O., Peterson, L. Reconstruction of the lateral ligaments of the ankle for chronic lateral instability. *J. Bone Joint Surg.* 70-A(4):581–588, 1988.
19. Karlsson, J., Bergsten, T., Lansinger, O., Peterson, L. Surgical treatment of chronic lateral instability of the ankle joint: a new procedure. *Am. J. Sports Med.* 17(2):268–273, 1989.
20. Chrisman, O.D., Snook, G. A. Reconstruction of lateral ligament tears of the ankle: an experimental study and clinical evaluation of seven patients treated by a new modification of the Elmslie procedure. *J. Bone Joint Surg.* 51-A:904–912, 1969.
21. Elmslie, R. C. Recurrent subluxation of the ankle joint. *Ann. Surg.* 110:364–367, 1934.
22. Evans, D. L. Recurrent instability of the ankle: a method of surgical treatment. *Proc. R. Soc. Med.* 46:343–344, 1953.
23. Lee, H. G. Surgical repair in recurrent dislocation of the ankle joint. *J. Bone Joint Surg.* 39-A:828–834, 1957.
24. Nilsson, H. Making a new ligament in ankle sprain. *J. Bone Joint Surg.* 31-A:380–381, 1949.
25. Karlsson, J., Bergsten, T., Lansinger, O., et al. Surgical treatment of chronic lateral instability of the ankle joint: a new procedure. *Am. J. Sports Med.* 17:268–273, 1989.
26. Inman, V. T. In *The Joints of the Ankle*, pp. 1–117, Williams & Wilkins, Baltimore, 1976.
27. Root, M. C., Weed, J. H., Orien, W. P. In *Normal and Abnormal Function of the Foot*, pp. 1–161, 295–346, Clinical Biomechanics Corporation, Los Angeles, 1977.
28. Freeman, M. A. Instabilities of the foot after lateral ligament injuries of the ankle. *J. Bone Joint Surg.* 47-B(4):669–677, 1965.
29. Konradsen, L., Raun, J. B. Ankle instability caused by delayed peroneal reaction time. *Acta Orthop. Scand.* 61:388–390, 1990.
30. Isakov, E., Mizrahi, J., Solzi, P., et al. Response of the peroneal muscles to sudden inversion of the ankle during standing. *Int. J. Sport Biomech.* 2:100–109, 1986.
31. Glencross, D., Thornton, E. Position sense following joint injury. *J. Sports Med. Phys. Fitness* 21:23–27, 1981.
32. Staples, O. S. Ruptures of the fibular collateral ligaments of the ankle. *J. Bone Joint Surg.* 57-A:101–107, 1975.
33. Dufek, J. S., Bates, B. T. Biomechanical factors associated with injury during landing during sports. *Sports Med.* 12:326–337, 1991.
34. Garrick, J. G. The frequency of injury, mechanism of injury, and epidemiology of ankle sprains. *Am. J. Sports Med.* 5:241–242, 1977.
35. Colville, M. R., Marder, R. A., Boyle, J. J., et al. Strain measurement in lateral ankle ligaments. *Am. J. Sports Med.* 18:196–200, 1990.
36. Ishii, T. S., Miyagawa, Fukubayashi, T., Hayashi, K. Subtalar stress radiography using dorsiflexion and supination. *J. Bone Joint Surg.* 78-B:56–60, 1996.
37. Louwerens, J. W., Ginai, A. Z., Van Linge, B., Snuders, C. J. Stress radiography of the talocrural and subtalar joints. *Foot Ankle Int.* 16:148–155, 1995.
38. Zell, B., Shereff, M. J., Greenspan, A., Liebowitz, S. Combined ankle and subtalar instability. *Bull. Hosp. J. Dis. Orthop. Inst.* 46:37–46, 1986.
39. Brantigan, J. W., Pedegana, L. R., Lippert, F. G. Instability of the subtalar joint: diagnosis by stress tomography in three cases. *J. Bone Joint Surg.* 59-A:321–324, 1977.
40. Meyer, J. M., Garcia, J., Hoffmeyer, P., Fritschy, D. The subtalar sprain: a roentgenographic study. *Clin. Orthop.* 226:169–173, 1986.
41. Hertel, J., Denegar, C. R., Monroe, M. M., Stokes, W. L. Talocrural and subtalar joint instability after lateral ankle sprain. *Med. Sc. Sports Exerc.* 31:1501–1508, 1999.
42. Fraser, G. H., Ahmed, A. M. Passive rotational stability of the weight-bearing talocrural joint: an in-vitro biomechanical study (abstract). *Orthop. Trans.* 7:248, 1983.
43. McCullough, C. J., Burge, P. D. Rotary stability of the load-bearing ankle: an experimental study. *J. Bone Joint Surg.* 62-B:460–464, 1980.
44. Stormont, D. M., Morrey, B. F., Kai-Nan, A., et al. Stability of the loaded ankle: relation between articular restraint and primary and secondary static restraint. *Am. J. Sports Med.* 13:295–300, 1985.
45. Cass, J. R., Settles, H. Ankle instability: in vitro kinematics in response to axial load. *Foot Ankle* 15:134–140, 1994.
46. Brunt, D., Anderson, J. C., Huntsman, B., et al. Postural responses to lateral perturbation in healthy subjects and ankle sprain patients. *Med. Sci. Sports Exerc.* 24:171–176, 1992.
47. Dietz, V., Berger, W. Spinal coordination of bilateral leg muscle activity during balancing. *Exp. Brain Res.* 47:172–176, 1982.
48. Nashner, L. M. Balance adjustments of humans perturbed while walking. *J. Neurophysiol.* 44:650–664, 1980.
49. Nashner, L. M., Black, F. O., Wall, C. Adaption to altered support and visual conditions during stance: patients with vestibular deficits. *J. Neurosci.* 2:536–544, 1982.
50. Nawoczenski, D. A., Owen, M. G., Ecker, M. L., et al. Objective evaluation of peroneal response to sudden inversion stress. *J. Orthop. Sports Phys. Ther.* 7:107–109, 1985.

51. Dietz, V. Human neuronal control of automatic functional movements: interaction between central programs and afferent input. *Physiol. Rev.* 72:33–42, 1992.
52. Dietz, V., Quintern, J., Sillem, M. Stumbling reactions in man: significance of proprioceptive and pre-programmed mechanisms. *J. Physiol.* 386:149–163, 1987.
53. Nashner, L. M. Adapting reflexes controlling the human posture. *Exp. Brain Res.* 26:59–72, 1976.
54. Mathews, P. B., Stein, R. B. The regularity of primary and secondary muscle spindle afferent discharges. *J. Physiol. Lond.* 202:59–82, 1969.
55. Robbins, S. E., Hanna, A. M., Gouw, G. J. Overload protection: avoidance response to heavy plantar surface loading. *Med. Sci. Sports Exerc.* 20:85–92, 1988.
56. Freeman, M. A. R., Dean, M. R. E., Hanham, I. W. F. The etiology and prevention of functional instability of the foot. *J. Bone Joint Surg.* 47-B:678–685, 1965.
57. DiCarlo, M. S., Talbot, R. W. Evaluation of ankle joint proprioception following injection of the anterior talofibular ligament. *J. Orthop. Sports Phys. Ther.* 8:70–76, 1986.
58. Feuerbach, M. A., Brabiner, M. O., Koh, T. J., Weiker, G. G. Effect of an ankle orthosis and ankle ligament anesthesia on ankle joint proprioception. *Am. J. Sports Med.* 20:223–229, 1994.
59. Barrack, R. L., Skinner, H. B., Brunet, M. E. Functional performance of the knee after Intra-articular anesthesia. *Am. J. Sports Med.* 11:258–261, 1983.
60. McCluster, G. M., Blackburn, T. A., Lewis, T. Prevention of ankle sprains. *Am. J. Sports Med.* 4:151–157, 1976.
61. Skinner, H. B., Wyatt, M. P., Hodgdon, D. W., Connerd, R. C. Effect of fatigue on joint position sense of the knee. *J. Orthop. Res.* 4:112–118, 1986.
62. Konradsen, L., Voight, M., Hojsgaard, C. Ankle inversion injuries: the role of the dynamic defense mechanism. *Am. J. Sports Med.* 25:54–58, 1997.
63. Karlsson, J., Andreasson, G. O. The effect of external ankle support in chronic lateral ankle joint instability: an electromyographic study. *Am. J. Sports Med.* 20:257–261, 1992.
64. Glick, J. M., Gordon, R. B., Nishimoto, D. The prevention and treatment of ankle injuries. *Am. J. Sports Med.* 4:136–141, 1976.
65. Manfroy, P. P., Ashton-Miller, J. A., Wojtys, E. M. The effect of exercise, prewrap and athletic tape on the maximal active and passive ankle resistance to ankle inversion. *Am. J. Sports Med.* 25:156–163, 1997.
66. Shapiro, M. S., Kabo, J. M., Mitchell, P. W., et al. Ankle sprain prophylaxis: an analysis of the stabilizing effects of bracing and tape. *Am. J. Sports Med.* 22:78–82, 1994.
67. Dietz, V., Schmidtbleicher, D., Noth, J. Neuronal mechanisms of human locomotion. *J. Neurophysiol.* 42:1212–1222, 1979.
68. Berger, W. G., Dietz, V., Quintern, J. Corrective reactions to stumbling reactions in man: neuronal coordination of bilateral leg muscle activity during gait. *J. Physiol. Lond.* 357:109–125, 1984.
69. Dietz, V., Schmidtbleicher, D. Interaction between pre-activity and stretch reflex in human triceps brachii during landing from forward falls. *J. Physiol.* 311:113–125, 1981.
70. Gottlieb, G., Agarwal, G. Response to sudden torques about the ankle in man. *J. Neurophysiol.* 42:91–106, 1979.
71. Nichols, T., Houk, J. Improvements in linearity and regulation of stiffness that results from action of stretch reflex. *J. Neurophysiol.* 39:119–142, 1976.
72. Gollhofer, A., Schmidtbleicher, D., Dietz, V. Regulation of muscle stiffness in human locomotion. *Int. J. Sports Med.* 5:19–22, 1984.
73. Armstrong, R. B. Initial events in exercise induced muscular injury. *Med. Sci. Sports Exerc.* 22:429–436, 1990.
74. Norman, R. W., Komi, P. V. Electromechanical delay in skeletal muscle under normal movement conditions. *Acta. Physiol. Scand.* 100:241–252, 1979.
75. Cavagna, G., Dunsman, B., Margaria, R. Positive work done by a previously stretched muscle. *J. Appl. Physiol.* 65:11–19, 1988.
76. Richie, D. H., Endo, C. K., DeVries, H. Shin muscle activity and sport surfaces: an electromyographic study. *J. Am. Podiatr. Med. Assoc.* 83:181–187, 1993.
77. Linge, B. V. Activity of peroneal muscles, the maintenance and balance, and the prevention of injury of the ankle: an electromyographic and kinematic study. *Acta. Orthop. Scand.* (suppl. 227):67–72, 1988.
78. Melvill-Jones, G., Watt, D. G. D. Muscular control of landing from unexpected falls in man. *J. Physiol.* 219:729–737, 1971.
79. Winter, D. A. *The Biomechanics and Motor Control of Human Gait: Normal, Elderly, and Pathological*, 2nd ed., p. 21, University of Waterloo Press, Waterloo, Ontario, Canada, 1991.
80. Ashton-Miller, J. A., Ottaviano, R. A., Hutchinson, C., Wojtys, E. M. What best protects the inverted ankle against further inversion? *Am. J. Sports Med.* 24:800–809, 1996.
81. Bullard, R. H., Dawson, J., Arenson, D. J. Taping the “athletic ankle.” *J. Am. Podiatr. Assoc.* 69:727, 1979.
82. Fumich, R. M., Ellis, A. E., Guerin, G. J., et al. The measured effect of taping on combined foot and ankle motion before and after exercise. *Am. J. Sports Med.* 9:165–170, 1981.
83. Larsen, E. Taping the ankle for chronic instability. *Acta. Orthop. Scand.* 55:551–553, 1984.
84. Quigley, T. B., Cox, J., Murphy, J. Protective wrapping for the ankle. *JAMA* 132:924, 1946.
85. Barrett, J. R., Tangi, J. L., Drake, C., et al. High versus low top shoes for the prevention of ankle sprains in basketball players: a prospective, randomized study. *Am. J. Sports Med.* 21:582–585, 1993.
86. Burks, R. T., Bean, B. G., Marcus, R., et al. Analysis of athletic performance with prophylactic ankle devices. *Am. J. Sports Med.* 19:104–106, 1991.
87. Journal, J. P. The effects of taping on vertical jumping ability. *J. Athletic Train.* 7:146–152, 1972.
88. Mayhew, J. L. Effects of ankle taping on motor performance. *J. Athletic Train.* 7:10–18, 1972.
89. Delacerda, F. G. Effect of underwrap conditions on the supportive effectiveness of ankle strapping with tape. *J. Sports Med. Phys. Fitness* 18:77–84, 1978.
90. Greene, T. A., Hillman, S. K. Comparison of support provided by a semirigid orthosis and adhesive ankle taping before, during and after exercise. *Am. J. Sports Med.* 18:498–506, 1990.
91. Gross, M. T., Bradshaw, M. K., Ventry, L. C., et al. Comparison of support provided by ankle taping and semirigid orthosis. *J. Orthop. Sports Phys. Ther.* 9:33–39, 1987.
92. Gross, M. T., Lapp, A. K., Davis, J. M. Comparison of Sweed-o universal ankle support and Aircast sport stirrup orthoses and ankle tape in restricting ankle eversion-inversion before and after exercise. *J. Orthop. Sports Phys. Ther.* 13:11–19, 1991.
93. Laughman, R. K., Carr, T. A., Chao, E. Y., et al. Three-dimensional kinematics of the taped ankle before and after exercise. *Am. J. Sports Med.* 8:425–431, 1980.
94. Bunch, R. P., Bednarski, K., Holland, D., et al. Ankle joint support: a comparison of reusable lace-on braces with taping and wrapping. *Physician Sports Med.* 13(5):59–62, 1985.
95. Malina, R. M., Plagenz, L. B., Rarick, G. L. Effect of exercise upon the measurable supporting strength of cloth and tape ankle wraps. *Res. Q.* 34:158–165, 1963.
96. Rarick, G. L., Bigley, G., Karst, R., et al. The measurable support of the ankle joint by conventional methods of taping. *J. Bone Joint Surg.* 44-A:1183–1190, 1962.

97. Rovere, G. D., Clarke, T. J., Yates, S. C., et al. Retrospective comparison of taping and ankle stabilizers in preventing ankle injuries. *Am. J. Sports Med.* 16:228–233, 1988.
98. Gross, M. T., Ballard, C. L., Mears, H. C., Watkins, E. J. Comparisons of Don-Joy ankle ligament protector and Aircast sport stirrup orthoses in restricting foot and ankle motion before and after exercise. *J. Orthop. Sports Phys. Ther.* 16:60–67, 1992.
99. Cornwall, M. W., Murrell, P. Postural sway following inversion sprain of the ankle. *J. Am. Podiatr. Med. Assoc.* 81:243–247, 1991.
100. Lofvengerg, R., Karrholm, J., Sudelin, G., Ahlgren, O. Prolonged reaction time in patients with chronic lateral instability of the ankle. *Am. J. Sports Med.* 23:414–417, 1995.
101. Irrgang, J. J., Whitney, S. L., Cox, E. D. Balance and proprioception training for rehabilitation of the lower extremity. *J. Sports Rehab.* 3:68–83, 1994.
102. Pinciuro, D., Lephart, S. M., Henry, T. Learning effects and reliability of the Biodex Stability System®. *J. Athletic Train.* 30: 535–541, 1995.
103. Tropp, H., Ekstrand, J., Gillquist, J. Stabilometry in functional instability of the ankle and its value in predicting injury. *Med. Sci. Sports* 16:64–66, 1984.
104. Lentell, G. L., Katzman, L. L., Walters, M. K. The relationship between muscle function and ankle stability. *J. Orthop. Sports Phys. Ther.* 11:605–611, 1990.
105. Hoffman, M., Payne, V. G. The effects of proprioceptive ankle disk training in healthy subjects. *J. Orthop. Sports Phys. Ther.* 21:90–93, 1995.
106. Tropp, H., Askling, C., Gillquist, J. Prevention of ankle sprains. *Am. J. Sports Med.* 13:259–262, 1985.
107. Tropp, H., Gillquist, J. Factors affecting stabilometry recordings of single limb stance. *Am. J. Sports Med.* 12:185–188, 1985.
108. Gauffin, H., Tropp, H., Oderrick, P. Effect of ankle disk training on postural control in patients with functional instability of the ankle joint. *Int. J. Sports Med.* 9:141–144, 1988.
109. Rozzi, S. L., Lephart, S. M., Sterne, R., Kuligowski, L. Balance training for persons with functionally unstable ankles. *J. Orthop. Sports Phys. Ther.* 8:478–486, 1999.
110. Bosien, W. R., Staples, O. S., Russell, S. W. Residual disability following acute ankle sprains. *J. Bone Joint Surg.* 37-A:1237–1243, 1955.
111. Staples, S. Ruptures of the fibular collateral ligaments of the ankle. *J. Bone Joint Surg.* 57:101–107, 1975.
112. Glasoe, W. M., Allen, M. K., Awtry, B. F., Yack, H. J. Weight bearing immobilization and early exercise treatment following a grade II lateral ankle sprain. *Orthop. Sports Phys. Ther.* 29:394–399, 1999.
113. Gauffin, H., Tropp, H., Odenrick, P. Effect of ankle disk training on postural control in patients with functional instability of the ankle joint. *Int. J. Sports Med.* 9:141–144, 1988.
114. Cooper, D., Farr, J. Ankle rehabilitation using the ankle disk. *Physician Sports Med.* 6:141, 1978.
115. Kaikkonen, A., Kannus, P., Jarvinen, M. A performance test protocol and scoring scale for the evaluation of ankle injuries. *Am. J. Sports Med.* 22(4):462–469, 1994.

Suggested Reading

- Bahr, R., Lian, O., Bahr, I. A. A twofold reduction in the incidence of ankle sprains in volleyball after the introduction of an injury prevention program: a prospective cohort study. *Scand. J. Med. Sci. Sports* 7(3):172, 1997.
- Brand, R. L., Black, H. M., Cox, J. S. The natural history of the inadequately treated ankle sprain. *Am. J. Sports Med.* 5:248–249, 1977.
- Brantigan, J. W., Pedegana, L. R., Lippert, F. G. Instability of the subtalar joint: diagnosis by stress tomography in three cases. *J. Bone Joint Surg.* 59-A:321–324, 1977.
- Bruns, J., Staerk, H. Mechanical ankle stabilization due to the use of orthotic devices and peroneal muscle strength an experimental investigation. *Int. J. Sports Med.* 13:611–615, 1992.
- Cawley, P. W., France, E. P. Biomechanics of the lateral ligaments of the ankle: an evaluation of the effects of axial load and single plane motions on ligament strain patterns. *Foot Ankle*, 12:92–99, 1991.
- Fraser, G. A., Ahmed, A. M. Passive rotational stability of the weight-bearing talocrural joint: an in vitro biomechanical study (Abstract). *Orthop. Trans.* 7:248, 1983.
- Friden, T., Zatterstrom, R., Lindstrand, A., Moritz, U. A stabilometric technique for evaluation of lower limb instability. *Am. J. Sports Med.* 17:118–122, 1989.
- Guskiewicz, K. M., Perrin, D. H. Effect of orthotics on postural sway following inversion ankle sprain. *J. Orthop. Sports Phys. Ther.* 23:326–331, 1996.
- Kelikian, H., Kelikian, A. S. *Disorders of the Ankle*, pp. 58–59, W.B. Saunders, Philadelphia, 1985.
- Lephart, S. M., Henry, T. J. Functional rehabilitation for the upper and lower extremity. *Orthop. Clin. North Am.* 26:579–592, 1995.
- Ramot, Y., Susak, Z. The dynamics of the subtalar joint in sudden inversion of the foot. *J. Biomech. Eng.* 112:9, 1990.
- Sheth, P., Yu, B., Laskowski, E. R., et al. Ankle disk training influences reaction times of selected muscles in a simulated ankle sprain. *Am. J. Sports Med.* 25:538, 1997.
- Verhagen, R. A., Kelzer, G., Van Dijk, C. N. Long-term follow-up of inversion trauma of the ankle. *Arch. Orthop. Trauma Surg.* 114:92–96, 1995.